

ALEXANDER R. VANCE

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PROFESSIONAL SUMMARY

Strategic and performance-driven Senior Backend Developer with over 10 years of experience in architecting, developing, and deploying high-scale distributed systems and microservices architectures. Expert in optimizing server-side logic to handle millions of concurrent requests while maintaining sub-millisecond latency through advanced caching strategies, database sharding, and asynchronous processing. Proven track record of leading cross-functional engineering teams through complex digital transformations, transitioning legacy monolithic applications into resilient, cloud-native environments using Go, Java, and Python. Specialized in building robust API ecosystems and data-intensive pipelines that prioritize security, observability, and cost-efficiency within AWS and GCP infrastructures. Committed to engineering excellence by implementing rigorous CI/CD standards and mentoring junior talent to foster a culture of technical rigor and continuous delivery.

TECHNICAL SKILLS

LANGUAGES: Go (Golang), Java (Spring Boot), Python (Django, FastAPI), Node.js (TypeScript), C++, SQL, Rust.

BACKEND FRAMEWORKS: Spring Framework, Gin, Echo, Flask, Express.js, NestJS, gRPC, Protobuf, GraphQL.

DATABASES & STORAGE: PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Elasticsearch, DynamoDB, Amazon S3.

CLOUD & INFRASTRUCTURE: AWS (EC2, Lambda, RDS, S3, EKS), Google Cloud Platform (GKE, Cloud Functions), Terraform, CloudFormation.

DEVOPS & TOOLS: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI, ArgoCD,

Prometheus, Grafana, ELK Stack.

MESSAGING & STREAMING: Apache Kafka, RabbitMQ, Amazon SQS/SNS, Google Pub/Sub, NATS.

ARCHITECTURE & TESTING: Microservices, Event-Driven Architecture, Serverless, Domain-Driven Design (DDD), JUnit, PyTest, Mocha, GoTest.

PROFESSIONAL EXPERIENCE

SCALETECH SYSTEMS | San Francisco, CA

Senior Staff Backend Engineer | January 2020 ? Present

- Spearheaded the architectural overhaul of a core payment processing engine, migrating a monolithic PHP backend to a high-concurrency Go-based microservices architecture, which resulted in a 45% increase in transaction throughput and a 30% reduction in server costs.
- Engineered a distributed locking mechanism using Redis and Redlock to prevent race conditions during high-volume promotional events, ensuring 100% data consistency across geographically distributed database clusters.
- Developed and maintained a robust set of internal gRPC-based APIs to facilitate low-latency communication between 50+ microservices, reducing internal network overhead by 25% compared to traditional RESTful implementations.
- Implemented a comprehensive observability suite using Prometheus and Grafana to monitor P99 latency and system health, allowing the team to proactively identify and resolve bottlenecks before they impacted end-users.
- Directed the integration of a multi-region Kafka event bus to decouple services and enable real-time data streaming for the analytics engine, processing over 2 billion events per day with zero data loss.
- Optimized PostgreSQL database performance by implementing sophisticated indexing strategies, query refactoring, and read-replica load balancing, which decreased average query response times from 200ms to 40ms.
- Mentored a team of 12 backend engineers through weekly code reviews and architecture design sessions, establishing a standardized "API-first" development culture that accelerated feature delivery cycles by 20%.

- Orchestrated the transition to a fully automated CI/CD pipeline using Terraform and GitHub Actions, enabling blue-green deployments that eliminated downtime during major production releases.
- Managed the security hardening of the backend infrastructure by implementing OAuth2/OIDC protocols and integrating HashiCorp Vault for centralized secret management and encryption at rest.
- Collaborated with product stakeholders to define technical roadmaps for the next generation of fintech products, ensuring that architectural decisions aligned with long-term business scalability goals.

DATASTREAM DYNAMICS | Austin, TX

Senior Backend Developer | June 2016 ? December 2019

- Designed and built a scalable ETL (Extract, Transform, Load) pipeline using Python and Apache Spark to ingest and process 5TB of daily telemetry data from IoT devices, enabling real-time business intelligence reporting.
- Authored a high-performance caching layer using Memcached and Redis that reduced database load by 60% during peak traffic periods, significantly improving application responsiveness for the global user base.
- Refactored legacy Java Spring Boot applications to utilize reactive programming patterns (Project Reactor), which enhanced the system's ability to handle concurrent I/O-bound tasks without thread starvation.
- Lead the containerization effort for the entire backend stack using Docker, creating standardized development environments that reduced "it works on my machine" issues by 90%.
- Integrated third-party APIs for international shipping and tax calculations, building a fault-tolerant circuit breaker pattern using Hystrix to ensure system stability when external dependencies failed.
- Developed a custom automated testing framework that increased code coverage from 40% to 85%, significantly reducing the number of regressions reaching the production environment.
- Optimized cloud resource utilization by implementing AWS Auto Scaling groups and transitioning non-critical batch jobs to AWS Lambda, resulting in an annual infrastructure savings of \$120,000.

- Collaborated with frontend teams to design and document comprehensive RESTful APIs using Swagger/OpenAPI, facilitating smoother integration and faster iteration on user-facing features.
- Implemented a role-based access control (RBAC) system to manage granular permissions for enterprise clients, ensuring compliance with SOC2 and GDPR data privacy regulations.
- Conducted deep-dive root cause analysis for complex production incidents, utilizing distributed tracing (Jaeger) to identify and rectify performance bottlenecks in the service mesh.

WEBLOGIC SOLUTIONS | Seattle, WA

Backend Developer | May 2012 ? May 2016

- Developed core server-side logic for a high-traffic e-commerce platform using Node.js and Express, supporting over 500,000 active monthly users.
- Designed the relational database schema for a complex inventory management system, utilizing normalization techniques to ensure data integrity and prevent redundancy across 100+ tables.
- Implemented an asynchronous notification system using RabbitMQ and Celery to handle email and SMS alerts, offloading heavy processing tasks from the main request-response cycle.
- Built a custom CMS backend that empowered non-technical staff to update site content dynamically, reducing the engineering team's workload for minor content changes by 40%.
- Collaborated with the DevOps team to migrate on-premise servers to AWS EC2 instances, facilitating easier scaling and improved disaster recovery capabilities.
- Created comprehensive unit and integration tests using Mocha and Chai, ensuring the reliability of critical business logic during a period of rapid feature expansion.
- Optimized front-end performance by implementing server-side rendering (SSR) logic and image optimization middleware, leading to a 1.5-second improvement in PageSpeed insights.
- Managed the version control strategy using Git, implementing a strict branching and merging policy that improved team collaboration and code quality.

TECHNICAL PROJECTS

REAL-TIME FINANCIAL CLEARING HOUSE ENGINE

- Architected a high-throughput transaction clearing system using Go and the Actor model to process financial orders with microsecond precision.
- Leveraged Apache Kafka for event sourcing, allowing the system to maintain a complete, immutable audit log of all financial movements for regulatory compliance.
- Implemented a custom sharding algorithm for the underlying CockroachDB cluster to ensure horizontal scalability as the volume of financial data grew.
- Developed a real-time fraud detection module using machine learning models served via a Python-based microservice, achieving a 99.8% accuracy rate in flagging suspicious activity.

GLOBAL MULTI-REGION DATABASE SYNCHRONIZER

- Engineered a synchronization tool to maintain data consistency across AWS regions (us-east-1 and eu-west-1) for a distributed PostgreSQL deployment.
- Utilized Change Data Capture (CDC) via Debezium and Kafka Connect to stream database mutations in real-time between disparate geographic locations.
- Developed a conflict resolution strategy based on Vector Clocks to handle concurrent writes to the same record, ensuring eventual consistency without manual intervention.
- Reduced cross-region data latency by 200ms by optimizing the serialization format of the synchronization payloads using Protocol Buffers (Protobuf).

DISTRIBUTED CACHING STRATEGY FOR E-COMMERCE

- Designed and implemented a multi-tier caching architecture involving local in-memory caches (LRU) and a centralized Redis cluster to handle 100k+ requests per second.
- Created a custom cache invalidation engine that utilized Pub/Sub messaging to purge stale data across all application nodes within 50ms of a database update.
- Improved the cache hit ratio from 65% to 92% by analyzing traffic patterns and implementing a predictive pre-fetching algorithm for popular product categories.
- Reduced the operational cost of the caching infrastructure by 20% by implementing data

compression for large JSON objects stored in Redis.

AI-DRIVEN LOG ANALYSIS AND ANOMALY DETECTION PIPELINE

- Developed a serverless data pipeline using AWS Lambda and Kinesis to ingest millions of log lines from various microservices in real-time.
- Integrated an ELK (Elasticsearch, Logstash, Kibana) stack to provide the engineering team with a centralized dashboard for searching and visualizing system errors.
- Built an anomaly detection service using Python's Scikit-learn that automatically alerted the on-call engineer when log patterns deviated from the historical baseline.
- Structured log data into a standardized JSON format across all services, improving the efficiency of automated log parsing and reducing storage requirements by 15%.

IOT TELEMETRY INGESTION AND PROCESSING LAYER

- Built a high-availability ingestion layer using Node.js and MQTT to handle persistent connections from over 500,000 IoT sensors simultaneously.
- Implemented a backpressure mechanism to prevent system saturation during spikes in telemetry data, utilizing a combination of buffer queues and rate-limiting.
- Designed a time-series data storage strategy using InfluxDB, allowing for high-speed writes and efficient aggregation of historical sensor data for trend analysis.
- Developed a web-socket based real-time monitoring portal that allowed users to visualize device status and sensor readings with less than 100ms of lag.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CERTIFICATIONS

- AWS Certified Solutions Architect ? Professional
- Google Professional Cloud Architect
- Certified Kubernetes Administrator (CKA)
- Oracle Certified Professional, Java SE Programmer